



AutoValuate
INTELLIGENCE

Explainable, damage-aware car valuation for the UAE.

One trained damage detector · one explainable price model · one agentic-RAG layer
— **every number citation-grounded.**

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Repo github.com/krish2105/AutoValuate-Intelligence

Live demo **deploying (runs locally today)**

The screenshot shows the AutoValuate web application interface. At the top, there's a header with the AutoValuate Intelligence logo and a status bar indicating 'API live'. Below the header, a tagline reads 'Know what your car is **really** worth.' followed by a subtext: 'An instant, explainable, damage-aware fair-market valuation for the UAE — backed by a trained damage detector, an explainable pricing model, and live comparable listings. Every number is traceable.' The main content area is divided into two columns. The left column, titled 'Your vehicle', contains a photo upload section with the text 'Drop car photos or tap to upload' and 'Up to 8 images - scanned for damage when the CV service is live'. Below this is a form with fields for Make (Toyota), Model (Corolla), Year (2019), Mileage (km) (98888), Cylinders (4), Body type (Sedan), Regional specs (GCC), Transmission (Automatic), Fuel (Petrol), City (Dubai), and Seller (Owner). A 'Value my car' button is at the bottom of the form. The right column, titled 'Your valuation appears here', contains a large empty box with a placeholder image and text: 'Fill in the details and hit "Value my car". The full reasoning trace streams live, then the results build in.'

THE PROBLEM

Everyone selling a car gets lowballed.

\$20.6B

UAE used-car market by 2026 — large, real, still growing

Dealers quote low

They quote below market to protect their own resale margin. You have no independent number to push back with.

Classifieds mislead

Listing sites show asking prices, not what cars actually sell for. Anchoring on them overprices your car.

Damage is guesswork

Both sides price that door dent by gut feeling. Nobody quantifies what a specific dent, at this mileage, actually costs.

WHY EXISTING TOOLS FALL
SHORT

Each piece exists. Nobody stitches them together.

Online valuers still tell you to pay for an in-person inspection to get a number you can trust. And not one consumer tool combines all three of these in a single report:



Visual damage assessment

Sees the dents, scratches, cracks — and localizes them.



Explainable pricing

Shows exactly why it reached a number, factor by factor.



Live comparable evidence

Backs the estimate with real listings you can open.

AutoValuate is the one report that carries all three — and ties every claim back to the model that produced it.

THE SOLUTION

Photos in. A trustworthy, explained number out.



No black box: the middle step is a purpose-trained CV model, a gradient-boosted price model with SHAP, and a hybrid retrieval layer — not one LLM guessing.

This is the running app.

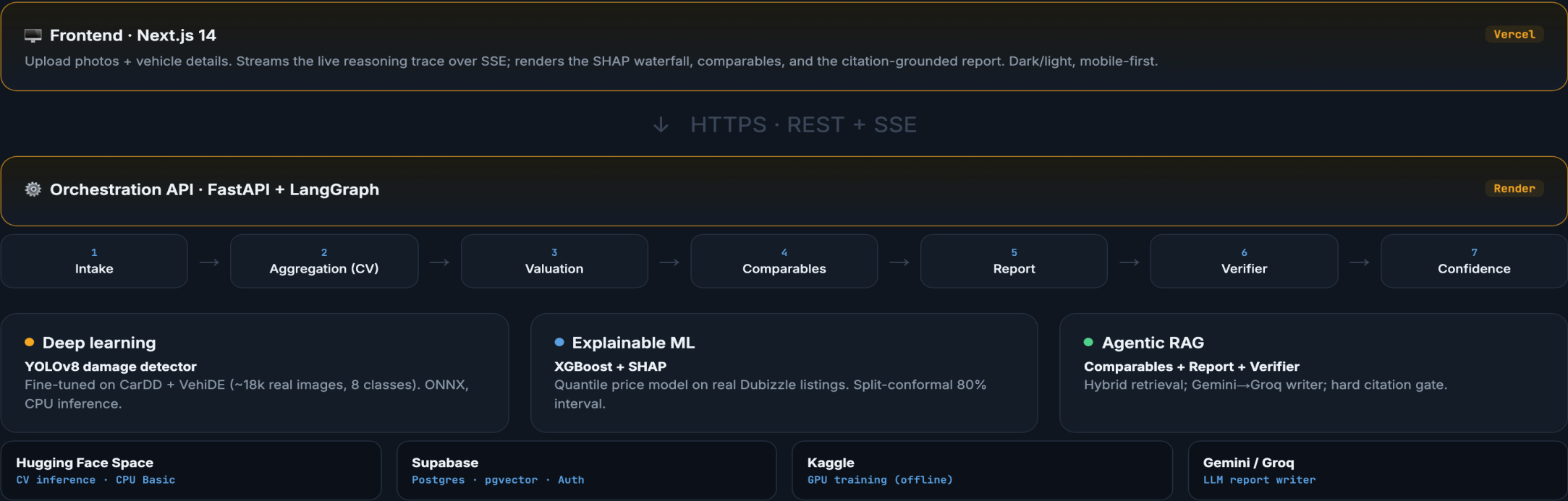
Runs locally today: **localhost:3000**

You're welcome to open the repo and run `./eval/run_all.sh` — every number in this deck reproduces.

A genuine hybrid — three AI systems, one product.

AutoValuate Intelligence — hybrid architecture

A trained deep-learning model + an explainable ML model + an agentic-RAG layer, on a fully free-tier stack.



A model trained to see damage — and where it is.

~18,000 real annotated images

CarDD + VehiDE, unified into one 8-class schema (dent, scratch, crack, glass shatter, lamp broken, tire flat, punctured, missing part).

YOLOv8 detection, not classification

Location matters for pricing — a windshield crack costs differently than a bumper scratch. So we detect and localize, not just label.

Verified pipeline

14,437 training + 1,184 validation images unified on Kaggle; ONNX export runs CPU-only on the free Hugging Face Space.



Honest numbers — the detector is still training.

mAP@0.5

lands here

held-out, per-class P-R from notebook 03

Why it's blank — and why that's the right call

The YOLOv8 fine-tune is running on Kaggle's free P100. I will not print a fabricated mAP. The eval harness already computes it on a strictly held-out split (no early-stop leakage), so the real number drops straight in when the run finishes.

What is already real and audited:

8-class unified dataset

14,437 train / 1,184 val — merged from CarDD + VehiDE, nothing dropped

Leakage-safe eval

held-out half of val by deterministic hash; ONNX-vs-PyTorch parity check

Deploy path proven

CPU ONNX service, full decode + NMS verified on synthetic output

A price you can interrogate, factor by factor.

19.6%

median error (held-out CV)

+28.4%

better than a naive baseline

0.80

calibrated price-interval coverage

3 / 3

SHAP directional checks pass

XGBoost quantile model on 672 real Dubizzle listings. SHAP explains each estimate; a split-conformal interval makes the stated 80% range mean 80%. Mileage and age push price down, newer year lifts it — exactly as economics demands.



Seven agents, streamed live — and a hard gate.

LangGraph orchestration

Intake → Aggregation (CV) → Valuation → Comparables → Report → Verifier → Confidence. Each step streams to the UI over SSE, so you watch the reasoning happen.

The Verifier — a real gate, not a promise

Every AED figure and citation in the report must trace to a computed value. In testing it caught an injected AED 999,999, a fake citation, and an ungrounded percentage.

Never blank

Gemini writes the report, Groq is the fallback, and a deterministic writer keeps it working with no keys at all.

The screenshot displays the AutoValuate Intelligence web application. The interface is divided into several sections:

- Header:** AutoValuate INTELLIGENCE logo and status indicators (API live, clock, refresh).
- Your vehicle:** A section for uploading photos or details. It includes a form with fields for Make (Toyota), Model (Corolla), Year (2019), Mileage (90000 km), Cylinders (4), Body type (Sedan), Regional specs (GCC), Transmission (Automatic), Fuel (Petrol), City (Dubai), and Seller (Owner). A "Value my car" button is at the bottom.
- Fair-market valuation:** Displays the estimated value of AED 33,184, calibrated with an 80% confidence range. It includes a bar chart showing the distribution of values from AED 21,894 to AED 50,294, with a median error of 19.57% on 672 real listings.
- Why this price — SHAP feature impact (AED):** A horizontal bar chart showing the impact of various features on the valuation, such as Engine size, Model year, Body type, Make, Km / year, and Seller.
- Live reasoning trace:** A section showing the reasoning process of the agents. It includes a "Live reasoning trace" header and a "Intake & validation" step with a green checkmark, indicating that the input (2019 Toyota Corolla, 90,000 km, 0 photo(s)) is valid.
- Visual damage assessment:** A section for computer-vision damage assessment, currently showing "No photo-based damage scan was performed for this valuation, so the estimate assumes market-typical condition. The trained YOLOv8".

It tells you how sure it is — and when not to trust it.

States its confidence

Every report gives the price-interval width and, when photos are provided, the detector's per-damage confidence.

Recommends inspection when unsure

Low or moderate confidence triggers a plain-English 'get a professional inspection' — it never presents a false-certain number.

Enforced, not aspirational

The Section-15 disclosure contract is a test: 90 checks across 18 vehicles, zero failures. It never claims to be a certified appraisal.

The screenshot displays the AutoValuate Intelligence interface. On the left, there's a form to input car details: Make (Lexus), Model (LX-Series), Year (2019), Mileage (90000 km), Cylinders (4), Body type (Sedan), Regional specs (GCC), Transmission (Automatic), Fuel (Petrol), City (Dubai), and Seller (Owner). A 'Value my car' button is at the bottom of the form.

On the right, a list of comparable vehicles is shown:

- 2021 Lexus Lx-Series: AED 329,000 (79,000 km, Dealer)
- 2019 Lexus Ux-Series: AED 85,000 (28,539 km, Dealer)
- 2019 Lexus Gx-Series: AED 135,000 (127,000 km, Owner)
- 2020 Lexus Es-Series: AED 92,000 (130,000 km, Owner)

Below the form, a 'Live reasoning trace' section lists the steps taken:

- Intake & validation: 2019 lexus lx-series, 90,000 km, 0 photo(s)
- Damage aggregation (CV): CV skipped (no CV service configured)
- Price model + SHAP: AED 46,439-106,675 (mid 70,384)
- Comparable listings (RAG): 5 comparable(s), top match 0.92
- Report synthesis: written via template
- Citation verifier: 13 numbers, 14 citations checked; all grounded

On the far right, a 'Seller report' section provides a synthesized valuation: 'Based on the details provided, 2019 Lexus Lx-Series has an estimated fair-market value between AED 46,439 and AED 106,675, with a mid-point of AED 70,384. This range is a calibrated 79% confidence interval. The main factors behind this estimate are make (+37,446 AED), year (-15,282 AED), noOfCylinders (-13,928 AED). On held-out testing the pricing model carries a median error of about 19.57%, so treat the mid-point as a guide, not a guarantee. A visual damage assessment was not available for this valuation, so the estimate assumes a market-typical condition — a professional inspection is recommended to confirm. Comparable live listings support this range: 2019 lexus lx-series 134000km at AED 240,000; 2021 lexus lx-series 79000km at AED 329,000; 2019 lexus ux-series 28539km at AED 85,000; 2019 lexus gx-series 127000km at AED 135,000; 2020 lexus es-series 130000km at AED 92,000. If the model's confidence is limited or the car has damage beyond what the photos show, a professional inspection is the safest next step before you set a

THE STACK · AND WHY IT'S
ALL FREE

Zero cost — with the engineering judgment to prove it.

Vercel

Frontend (Next.js)

Render

Orchestration API

Hugging Face

CV inference · CPU Basic

Supabase

Postgres · pgvector · Auth

Kaggle

GPU training (offline)

Gemini / Groq

LLM report writer

Render, not Railway (Railway has no real free tier in 2026). **The CV model runs on Hugging Face** — 512MB can't hold it. **A scheduled ping** keeps Supabase from auto-pausing, so the link never dies.

EVALUATION SUMMARY

Every number here reproduces from the repo.

19.6%

valuation median error

1.00

comparables same-make P@5

1.000

report faithfulness

0.000

faithfulness — hallucinated control

90 / 90

confidence-contract checks

53

integration checks, all green

CV mAP@0.5 lands here once training finishes — the same held-out harness, no placeholder in its place.

What it does not do — named on purpose.

CV detector still training

The mAP isn't in yet; the deep-learning claim rests on the finished run, and I'd rather show that than a guess.

Modest tabular set

672 real listings is small, so the price interval is genuinely wide — which is exactly why the range is disclosed, not hidden.

CPU-speed inference

The free tier runs the detector on CPU: a few seconds per image, batched — fine for a demo, not high-throughput.

No accident-history feed

There's no free UAE accident-record API, so the estimate can't see undisclosed history — hence 'get an inspection.'

Where it goes next.

01

Dealer bulk API

Batch intake + a verified-estimate API for used-car dealers — the paying tier.

02

Dubizzle auto-fill

A browser extension that pre-fills a listing straight from a completed valuation.

03

Price-trend alerts

'Your car's segment dropped 4% this month' — turning a one-shot tool into a reason to return.

All post-MVP — clearly future work, not claimed as built.

One project, three things proven.

- **Deep learning** a trained, evaluated CV damage detector
- **Explainable ML** XGBoost + SHAP, a price you can interrogate
- **Agentic RAG** orchestration with a hard citation gate

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Live demo [deploying \(runs locally today\)](#)

Thank you — happy to open any part of it live.